

Theorem C is a corollary of Rick's Theorem 2.5,
and the derivation is a telescoping with one broken step

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Abstract

I close `gap:rick` of `2026-10-07-two-part-green-polynomials.tex`. Rick's Theorem 2.5 (the *Two-point formula*, labelled `thm:2pt` in his FPSAC 2027 draft) is a closed form for $\langle T_a g, p_x p_y \rangle$ for *every* symmetric g in a variables; transported by his dictionary it therefore computes $Y_{(x,y)}^\lambda(t)$ for *every* λ . My Theorem C is its specialisation to $\ell(\lambda) = 2$, and I prove the specialisation: his formula, read at $a = 2$, is a linear functional on the Laurent coefficients of $G_1(w) = P_\rho(1, w; t)$ which *telescopes* against the potential $\alpha(k) = (t^{-k} + t^{k-1})/(1+t)$ at every index except $e = 0$, and the single broken step is exactly the source of the integer terms in Theorem C. So the brief's outcome 2 holds: A = B, **after a rewrite that is not a substitution**. The rewrite is verified symbolically in $\mathbb{Q}(t)$ on all 271 ordered rows with $|\lambda| \leq 12$, and against two ground truths — one of which uses no tableaux at all — with 0 disagreements. Consequences I volunteer: Theorem C is **demoted to a corollary** with an independent proof; Theorem B is *not* a corollary, because it is a statement about the h -expansion rather than about values; and Theorem B composed with Theorem 2.5 gives a closed form for $[h_\nu]Q'_\lambda$ for *all* λ and all $\ell(\nu) \leq 2$, which is the item my own paper listed as not attempted.

1 Conventions

All conventions are those of `2026-10-07-two-part-green-polynomials.tex`: the ordinary Hall pairing $\langle \cdot, \cdot \rangle$ with $\langle p_\lambda, p_\mu \rangle = \delta_{\lambda\mu} z_\lambda$; the Hall-Littlewood pairing $\langle p_\lambda, p_\mu \rangle_t = \delta_{\lambda\mu} z_\lambda \prod_i (1 - t^{\lambda_i})^{-1}$; $\varphi_m(t) = \prod_{k=1}^m (1 - t^k)$; $b_\mu(t) = \prod_{i \geq 1} \varphi_{m_i(\mu)}(t)$; $n(\mu) = \sum_i (i-1)\mu_i$; $Q'_\mu = \sum_\nu K_{\nu\mu}(t) s_\nu$ with K in the charge normalisation; and

$$Y_\rho^\lambda(t) = \langle p_\rho, Q'_\lambda \rangle = [P_\lambda] p_\rho = \sum_\nu K_{\nu\lambda}(t) \chi_\rho^\nu. \quad (1)$$

Rick's X_μ^λ of $p_\mu = \sum_\lambda X_\mu^\lambda(t) P_\lambda$ is literally Y_μ^λ ; this needs no computation, since that equation pins X with no normalisation freedom. Throughout, $\lambda = (a, b)$ with $a \geq b \geq 1$, $n = a + b$, $\rho = (a-1, b-1)$, $m = a - b$, and $\mu = \{x, y\}$ is a two-part class of n . I write $\llbracket \cdot \rrbracket$ for an indicator.

Theorem 1 (C, restated from the earlier paper). *With $m_0 = \min(x, y)$,*

$$Y_{\{x,y\}}^{(a,b)}(t) = \begin{cases} 1 + \llbracket a = b \rrbracket + (t-1)t^{b-1}, & m_0 = b, \\ (t-1)t^{b-1} + (t-1)t^{b-m_0-1}, & 1 \leq m_0 \leq b-1, \\ (t-1)t^{b-1}, & m_0 > b. \end{cases}$$

2 gap:rick closed: the statement was on disk under a different label

gap:rick read: “the statement itself is in grandpa-rick/work-in-progress at 71b4cad, which I do not hold.” I hold it, and have since 2026-10-06. It is at

/tmp/rick-review/fpsac2027/fpsac2027-draft.tex, environment thm:2pt, *Two-point formula*.

The string 2.5 does not occur anywhere near it. The identification is made by the *registry comment line immediately above the environment*,

```
% registry: two-point-string-formula-thm25 (+ shuffle-identity-lemma24), grade:
               proved
```

together with /tmp/rick-review/registry/hikita-star-dominance-support.json, whose node two-point-string-formula-thm25 carries approach: “Thm 2.5: closed $\langle T_ag, p_x p_y \rangle$ for all symmetric g .” Three further corroborations, each independent of that one:

- (i) the same node’s novelty_wake226 field states the dictionary $\Phi_a(P_\rho; x, y) = (1 - t^x)(1 - t^y)X_{(x,y)}^\lambda / b_\lambda$ with $\lambda = \rho + 1^a$, which is the dictionary his 2026-10-07 note attributes to Theorem 2.5;
- (ii) his §3(b) substitution in that note is reproduced *term for term* by setting $a = 2$ in thm:2pt (§3.1 below);
- (iii) his prose description of Theorem 2.5 — “explicit, with no sum over classes, linear in the two-string specialisation G_A of P_ρ ; not a product formula” — is the shape of thm:2pt and of nothing else in the draft.

Remark 2 (the failure mode, named). This is a third distinct way a holdings record can be wrong, and the sharpest so far. The two I had recorded were *a filename is a claim about contents* and *a scratch directory is invisible to an audit that only scans designated corpora*. This one is: **a theorem number is a claim about a numbering, not about contents**. I searched for “Theorem 2.5” in a document that numbers its theorems by L^AT_EX label and section, and concluded absence. The instrument that resolved it was his own % registry: comment — a channel I had read past for two sessions, in the same file, four lines above the thing I said I did not hold.

3 Rick’s Theorem 2.5, verbatim

From his Prop. 2.1 (subset formula) notation: for $F \in \Lambda_N$, $c_A = \prod_{i \in A, j \notin A} \frac{x_i - t x_j}{x_i - x_j}$ and $X_A = \prod_{i \in A} x_i$; and $T_ag := \sum_{|A|=a} c_A X_A g(X_A)$ for $g \in \Lambda_a$. His Notation paragraph declares $\langle \cdot, \cdot \rangle$ to be the *Hall pairing* and $\langle \cdot, \cdot \rangle_t$ the HL pairing, so the bracket in the statement below is unambiguous.

Theorem 3 (Rick, Thm. 2.5 = thm:2pt; quoted). *Let $g \in \Lambda_a$ be homogeneous of degree d , $n = a + d$, $x, y \geq 1$, $x + y = n$, and $\Phi_a(g; x, y) := \langle T_ag, p_x p_y \rangle$. For $1 \leq A \leq a - 1$ put $B = a - A$*

and $G_A(w) = \sum_k G_{A,k} w^k := g(1, t, \dots, t^{A-1}, w, wt, \dots, wt^{B-1})$. Then

$$\Phi_a(g; x, y) = (-1)^a (1-t^x)(1-t^y) \left\{ \sum_{A=1}^{a-1} t^{-AB} \left[\frac{G_{A,y-B}}{(1-t^A)(1-t^B)} + \frac{1}{1-t^a} \sum_{m \geq 1} (t^{-Am} - t^{Bm}) G_{A,y-B-m} \right] - \frac{g(1, t, \dots, t^{a-1})}{1-t^a} \sum_{j=0}^{a-1} t^{-jy} \right\}.$$

Lemma 4 (his dictionary = my `lem:dict`). For $\lambda = \rho + 1^a$ with $a = \ell(\lambda)$, $\Phi_a(P_\rho; x, y) = (1-t^x)(1-t^y) \frac{Y_{(x,y)}^\lambda(t)}{b_\lambda(t)}$.

Two things about the quantifiers in Theorem 3, both of which I checked by reading the statement rather than inferring from its use. First, g is an *arbitrary* homogeneous element of Λ_a . Second, $\lambda = \rho + 1^a$ with $a = \ell(\lambda)$ places *no* restriction on λ : every partition is $\rho + 1^{\ell(\lambda)}$ for $\rho = \lambda - 1^{\ell(\lambda)}$. Hence:

Verdict 5 (Task 3, first half). Theorem 3 together with Lemma 4 is a closed form for the *entire* two-part-class slice $\{Y_{(x,y)}^\lambda : \lambda \vdash n, x + y = n\}$, for all λ , not only two-row λ . The slice of my Theorem B is *inside* his, and the slice of my Theorem C is inside that.

Verdict 5 is a statement about the hypotheses, but I also tested it: his closed form, coded verbatim for general a , agrees with ground truth on all $\lambda \vdash n$ for $n \leq 8$ — $\ell(\lambda) = 1, \dots, 8$ — in 2100 comparisons, 0 disagreements, broken down by length in §6.

3.1 His $a = 2$ substitution is a correct specialisation of his own theorem

Set $a = 2$ in Theorem 3. Then $A = B = 1$ is the only term; $t^{-AB} = t^{-1}$; $(1-t^A)(1-t^B) = (1-t)^2$; $1-t^a = 1-t^2$; $\sum_{j=0}^1 t^{-jy} = 1+t^{-y}$; $(-1)^a = +1$; and $G_1(w) = g(1, w)$. Every one of these matches his §3(b) expression character for character. Outcome 3 of the brief (“his substitution is wrong”) is therefore excluded *by inspection*, before any numerical work. I nevertheless carry his §3(b) expression as a separate column (B') throughout, transcribed from his note and not re-derived, so that the claim is tested and not merely asserted.

For $g = P_\rho$ with $\rho = (a-1, b-1)$ in two variables,

$$G_1(w) = P_\rho(1, w; t) = w^{b-1} \left(1 + w^m + (1-t) \sum_{0 < k < m} w^k \right), \quad G_1 = w^{b-1} \text{ if } m = 0, \quad (2)$$

so the exponent support of G_1 is the integer interval $[b-1, a-1]$ with coefficient 1 at the two endpoints and $1-t$ strictly inside; and

$$G_1(t) = P_\rho(1, t; t) = \begin{cases} t^{b-1}(1+t), & a > b, \\ t^{b-1}, & a = b, \end{cases} \quad (3)$$

because for $m \geq 1$, $1 + t^m + (1-t)(t + \dots + t^{m-1}) = 1 + t^m + (t - t^m) = 1 + t$. (Both (2) and (3) are the $n = 2$ case of Macdonald’s principal specialisation, but neither needs it; I do not hold the book and quote no equation number from it.)

4 The specialisation, proved

4.1 A linear functional

For a Laurent polynomial $F(w) = \sum_k F_k w^k$ put

$$L_y(F) := t^{-1} \left[\frac{F_{y-1}}{(1-t)^2} + \frac{1}{1-t^2} \sum_{j \geq 1} (t^{-j} - t^j) F_{y-1-j} \right] - \frac{F(t)(1+t^{-y})}{1-t^2}, \quad (4)$$

so that, by §3.1 and Lemma 4,

$$Y_{(x,y)}^{(a,b)}(t) = b_{(a,b)}(t) \cdot L_y(G_1). \quad (5)$$

L_y is linear, so evaluate it on a monomial. Put $e := y - 1 - c$. Then $F_{y-1} = \llbracket e = 0 \rrbracket$ and the j -sum retains only $j = e$, and only when $e \geq 1$:

$$L_y(w^c) = \Psi(e) - \frac{t^c(1+t^{-y})}{1-t^2}, \quad \Psi(e) := \frac{t^{-1} \llbracket e = 0 \rrbracket}{(1-t)^2} + \frac{t^{-1}(t^{-e} - t^e) \llbracket e \geq 1 \rrbracket}{1-t^2}, \quad (6)$$

with $\Psi(e) = 0$ for $e \leq -1$. **Everything turns on the sign of e :** $e \geq 0$ iff $c \leq y - 1$, so his formula sees only the part of G_1 at or below w^{y-1} . That one-sided truncation is the whole of the apparent $x \leftrightarrow y$ asymmetry of Theorem 3.

4.2 The potential

Put $\psi(e) := (1-t)^2 \Psi(e)$, so $\psi(e) = 0$ for $e < 0$, $\psi(0) = t^{-1}$, and

$$\psi(e) = \frac{(1-t)(t^{-e} - t^e)}{t(1+t)} = (1-t)^2 t^{-1-e} (1 + t^2 + \dots + t^{2e-2}) \quad (e \geq 1).$$

Define

$$\boxed{\alpha(k) := \frac{t^{-k} + t^{k-1}}{1+t}} \quad (k \in \mathbb{Z}). \quad (7)$$

$\alpha(k)$ is a Laurent polynomial: for $k \geq 1$ the numerator is $t^{-k}(1 + t^{2k-1})$ and $2k - 1$ is odd, so $1 + t$ divides it; for $k \leq 0$ likewise. Three identities, each one line:

$$(T1) \quad \psi(e) = \alpha(e+1) - \alpha(e) \quad (e \geq 1), \quad (8)$$

$$(T2) \quad t\alpha(k+1) - \alpha(k) = (t-1)t^{k-1}, \quad \alpha(k+1) - t\alpha(k) = (1-t)t^{-k-1} \quad (k \in \mathbb{Z}), \quad (9)$$

$$(T3) \quad \alpha(0) = \alpha(1) = t^{-1}. \quad (10)$$

Proof. (T1): both sides equal $(1-t)^2 t^{-1-e} (1-t^{2e}) / (1-t^2)$, since $\alpha(e+1) - \alpha(e) = \frac{t^{-e-1} + t^e - t^{-e} - t^{e-1}}{1+t} = \frac{t^{-e-1}(1-t)(1-t^{2e+1}/t^2)}{1+t}$ — more directly, clearing $1+t$, $(1+t)(\alpha(e+1) - \alpha(e)) = t^{-e-1} - t^{-e} + t^e - t^{e-1} = -t^{-e-1}(t-1) - t^{e-1}(1-t) = (1-t)(t^{-e-1} - t^{e-1})$, and $(1+t)\psi(e) = (1-t)t^{-1}(t^{-e} - t^e) = (1-t)t(t^{-e-1} - t^{e-1})$. (T2): $(1+t)(t\alpha(k+1) - \alpha(k)) = t^{-k} + t^{k+1} - t^{-k} - t^{k-1} = t^{k-1}(t^2 - 1)$, and dividing by $1+t$ gives $-t^{k-1}(1-t)$; similarly $(1+t)(\alpha(k+1) - t\alpha(k)) = t^{-k-1} + t^k - t^{-k-1} - t^k = t^{-k-1}(1-t^2)$. (T3): $\alpha(0) = (1+t^{-1})/(1+t) = t^{-1}$ and $\alpha(1) = (t^{-1} + 1)/(1+t) = t^{-1}$. $\square$

Remark 6 (the broken step, which is the content). By (T3) the telescoped increment at $e = 0$ is $\alpha(1) - \alpha(0) = 0$, whereas $\psi(0) = t^{-1} \neq 0$. So ψ **telescopes against α at every index $e \geq 1$ and fails to telescope at exactly one index, $e = 0$** ; it also fails for $e \leq -1$, where $\psi = 0$ but $\alpha(0) - \alpha(-1) = -t^{-2}(1-t)^2 \neq 0$. The lone surviving t^{-1} at $e = 0$ is what produces the integer terms $1, 1 + \llbracket a = b \rrbracket, 2$ of Theorem 1; every other term of Theorem 1 is a telescoping boundary value. This is the whole mechanism, and it is ablated in §7: replacing ψ by the naive telescoped increment at all e drops agreement from 110/110 to 20/110.

4.3 Assembly

Put $E_1 := y - a$ and $E_2 := y - b$, so $E_2 - E_1 = m$, $-E_2 - 1 = b - y - 1$, and $-E_1 = x - b$. Since $x, y \geq 1$ we have $E_1 \leq b - 1$ and $E_2 \geq 1 - b$.

Case $a > b$. Here $b_\lambda = \varphi_1^2 = (1-t)^2$ and $(1-t)^2/(1-t^2) = (1-t)/(1+t)$, so by (3) the subtracted term of (4), scaled by b_λ , is exactly $-(1-t)t^{b-1}(1+t^{-y})$. Summing (6) over the monomials of G_1 — endpoints $c = b - 1, a - 1$, i.e. $e = E_2, E_1$; interior $c \in [b, a - 2]$, i.e. $E_1 < e < E_2$ — gives

$$Y_{(x,y)}^{(a,b)} = \psi(E_1) + \psi(E_2) + (1-t) \sum_{E_1 < e < E_2} \psi(e) - (1-t)t^{b-1}(1+t^{-y}). \quad (\star)$$

Note that $(\star)$ reproduces the shape of G_1 itself: endpoints weight 1, interior weight $1 - t$. Five subcases exhaust the signs, and $E_1 = E_2 = 0$ cannot occur because $m \geq 1$.

- (i) $E_1 \geq 1$ (so $E_2 \geq 2$). Then $y \geq a + 1$, hence $x \leq b - 1$ and $m_0 = x$. All indices are ≥ 1 , so $\sum_{E_1 < e < E_2} \psi = \alpha(E_2) - \alpha(E_1 + 1)$ by (T1) — correct also when $m = 1$, where the sum is empty and $E_2 = E_1 + 1$. With (T1) the brace of $(\star)$ becomes

$$[t\alpha(E_1+1) - \alpha(E_1)] + [\alpha(E_2+1) - t\alpha(E_2)] \stackrel{(T2)}{=} (t-1)t^{E_1-1} + (1-t)t^{-E_2-1}.$$

The second term is $(1-t)t^{b-y-1}$ and **cancels** against $-(1-t)t^{b-1-y}$ in $(\star)$. With $E_1 - 1 = b - x - 1$ this leaves $(t-1)t^{b-1} + (t-1)t^{b-m_0-1}$: Theorem 1, branch $1 \leq m_0 \leq b - 1$.

- (ii) $E_1 \leq -1, E_2 \geq 1$. Then $y \geq b + 1$ and $x \geq b + 1$, so $m_0 > b$. $\psi(E_1) = 0$; the interior contributes $\psi(0) = t^{-1}$ plus $\alpha(E_2) - \alpha(1)$, and by (T3) the t^{-1} **cancels** $\alpha(1)$, leaving $\alpha(E_2)$. The brace is $\alpha(E_2+1) - t\alpha(E_2) = (1-t)t^{b-y-1}$, which cancels as before. Hence $Y = (t-1)t^{b-1}$: branch $m_0 > b$.
- (iii) $E_2 = 0$, i.e. $y = b, x = a, m_0 = b$. Only $\psi(0) = t^{-1}$ survives, and $-(1-t)t^{b-1}t^{-b} = -(1-t)t^{-1} = -t^{-1} + 1$ absorbs it. Hence $Y = 1 + (t-1)t^{b-1}$: branch $m_0 = b$ with $\llbracket a = b \rrbracket = 0$.
- (iv) $E_2 \leq -1$, i.e. $y \leq b - 1, m_0 = y$. Every ψ vanishes and $Y = (t-1)t^{b-1} + (t-1)t^{b-y-1}$: branch $1 \leq m_0 \leq b - 1$.
- (v) $E_1 = 0$, i.e. $y = a, x = b, m_0 = b, E_2 = m \geq 1$. The brace is $t^{-1} + \alpha(E_2+1) - t\alpha(E_2) - (1-t)\alpha(1)$, and (T3) turns the last term into $-(1-t)t^{-1} = -t^{-1} + 1$; the remaining $(1-t)t^{b-y-1}$ cancels. Hence $Y = 1 + (t-1)t^{b-1}$: branch $m_0 = b$.

Case $a = b$. Now $G_1 = w^{b-1}$ alone, $b_\lambda = \varphi_2 = (1-t)(1-t^2)$, so $b_\lambda\Psi(e) = (1+t)\psi(e)$, $G_1(t) = t^{b-1}$, and with $E := y - b$,

$$Y_{(x,y)}^{(b,b)} = (1+t)\psi(E) - (1-t)t^{b-1}(1+t^{-y}).$$

Three subcases, and $m_0 > b$ is impossible because $m_0 \leq n/2 = b$:

- $E = 0$ ($x = y = b$): $(1+t)t^{-1} = t^{-1} + 1$ and $-(1-t)t^{-1} = -t^{-1} + 1$, so $Y = 2 + (t-1)t^{b-1}$. This is the *only* place the $\llbracket a = b \rrbracket$ term of Theorem 1 is exercised, and it arises from $b_\lambda = \varphi_2$ rather than φ_1^2 — i.e. from the dictionary, not from the formula.
- $E \leq -1$ ($y < b$, $m_0 = y$): $\psi = 0$ and $Y = (t-1)t^{b-1} + (t-1)t^{b-y-1}$.
- $E \geq 1$ ($y > b$, $m_0 = x = 2b - y$): $(1+t)\alpha(k) = t^{-k} + t^{k-1}$, so $(1+t)\psi(E) = t^{b-y-1} + t^{y-b} - t^{b-y} - t^{y-b-1}$; the terms t^{b-y-1} and t^{b-y} cancel against $-(1-t)t^{b-1-y}$, leaving $Y = (t-1)t^{b-1} + (t-1)t^{y-b-1}$ with $y - b - 1 = b - m_0 - 1$.

Theorem 7 (the specialisation). *For every $a \geq b \geq 1$ and every $x, y \geq 1$ with $x + y = a + b$, Rick’s Theorem 3 at $a = 2$ with $g = P_{(a-1, b-1)}$, transported by Lemma 4, equals the right-hand side of Theorem 1, as an identity in $\mathbb{Q}(t)$. Consequently Theorem C is a corollary of Theorem 2.5, and — since the derivation above is reversible, being a chain of identities — the independent proof of Theorem C in the earlier paper is a proof of Theorem 2.5 restricted to $a = 2$, $g = P_\rho$.*

Verdict 8 (Task 2). **Outcome 2 of the brief:** $A = B$, after a rewrite that is not a substitution. The transition map is exhibited: L_y of (4) read monomial-wise on G_1 , telescoped against α of (7), with the one broken index of Remark 6.

Remark 9 (what the rewrite actually buys, stated without inflation). Theorem 1 has three features that Theorem 3 does not display and that the telescoping is what makes visible.

- (1) *Manifest $x \leftrightarrow y$ symmetry.* $\Phi_a(g; x, y)$ is symmetric by definition, but the closed form of Theorem 3 is written one-sidedly in y ; at $a = 2$ the symmetry holds only after the cancellations above. (Checked: both orderings of every μ are separate rows of the table.)
- (2) *Manifest Laurent-polynomiality.* Theorem 3 at $a = 2$ is a sum of terms with poles at $t = 0, \pm 1$; the output has none.
- (3) *Independence of a except through $\llbracket a = b \rrbracket$.* The a -dependence of Theorem 3’s input sits in the upper endpoint $a - 1$ of G_1 ’s support and in (3); the telescoping is precisely what makes the upper-endpoint contribution cancel.

These are properties of a normal form, not new values. I claim no more for them than that.

5 The table

Five columns were computed, each from a different mechanism:

col	what	mechanism
A	Theorem 1, closed form	charge $\rightarrow$ <code>lem:mono</code> $\rightarrow$ <code>prop:tworow</code> $\rightarrow$ Thm B $\rightarrow$ Thm C
A'	Theorem 1, indicator form	a second transcription of the same statement
B	Theorem 3, coded verbatim for general a	his t -strings + Lemma 4
B'	his §3(b) $a = 2$ substitution, transcribed	his substitution + Lemma 4
C1	$\sum_\nu K_{\nu\lambda}(t)\chi_\rho^\nu$	charge Kostka–Foulkes $\times$ Murnaghan–Nakayama
C2	$b_\lambda \langle p_\rho, P_\lambda \rangle / \prod_i (1 - t^{\rho_i})$	Gram–Schmidt in the p -basis + plethysm. No tableaux.

A versus C2 is the only pair that shares no machinery: A reaches the value through charge and the h -expansion, C2 through orthogonalisation for $\langle \cdot, \cdot \rangle_t$ and the plethystic definition of Q' . A versus C1 shares the charge statistic, so it is a weaker control and is reported separately. B versus C2 shares Lemma 4 with B; §6 decouples that.

There were no disagreements, at any row, in any pair, at any value of t . Since there is no witness to report, what follows is the common value, and then — in §6–§7 — the counts, the stratum split, and the rank line, which are the parts of a green table that carry information.

λ	μ	branch	stratum	common value of the five columns
$n = 2$				
(1, 1)	(1, 1)	$m=b$	hook, $a=b$, $x=y$, $\min=1$	$t + 1$
$n = 3$				
(2, 1)	(2, 1)	$m=b$	hook, $\min=1$	t
$n = 4$				
(3, 1)	(3, 1)	$m=b$	hook, $\min=1$	t
(3, 1)	(2, 2)	$m>b$	hook, $x=y$	$t - 1$
(2, 2)	(3, 1)	$m<b$	$a=b$, $\min=1$	$(t - 1)(t + 1)$
(2, 2)	(2, 2)	$m=b$	$a=b$, $x=y$	$t^2 - t + 2$
$n = 5$				
(4, 1)	(4, 1)	$m=b$	hook, $\min=1$	t
(4, 1)	(3, 2)	$m>b$	hook	$t - 1$
(3, 2)	(4, 1)	$m<b$	$\min=1$	$(t - 1)(t + 1)$
(3, 2)	(3, 2)	$m=b$	<i>generic</i>	$t^2 - t + 1$
$n = 6$				
(5, 1)	(5, 1)	$m=b$	hook, $\min=1$	t
(5, 1)	(4, 2)	$m>b$	hook	$t - 1$
(5, 1)	(3, 3)	$m>b$	hook, $x=y$	$t - 1$
(4, 2)	(5, 1)	$m<b$	$\min=1$	$(t - 1)(t + 1)$
(4, 2)	(4, 2)	$m=b$	<i>generic</i>	$t^2 - t + 1$
(4, 2)	(3, 3)	$m>b$	$x=y$	$t(t - 1)$
(3, 3)	(5, 1)	$m<b$	$a=b$, $\min=1$	$t(t - 1)(t + 1)$
(3, 3)	(4, 2)	$m<b$	$a=b$	$(t - 1)(t^2 + 1)$
(3, 3)	(3, 3)	$m=b$	$a=b$, $x=y$	$(t + 1)(t^2 - 2t + 2)$
$n = 7$				
(6, 1)	(6, 1)	$m=b$	hook, $\min=1$	t
(6, 1)	(5, 2)	$m>b$	hook	$t - 1$
(6, 1)	(4, 3)	$m>b$	hook	$t - 1$
(5, 2)	(6, 1)	$m<b$	$\min=1$	$(t - 1)(t + 1)$
(5, 2)	(5, 2)	$m=b$	<i>generic</i>	$t^2 - t + 1$
(5, 2)	(4, 3)	$m>b$	<i>generic</i>	$t(t - 1)$
(4, 3)	(6, 1)	$m<b$	$\min=1$	$t(t - 1)(t + 1)$
(4, 3)	(5, 2)	$m<b$	<i>generic</i>	$(t - 1)(t^2 + 1)$
(4, 3)	(4, 3)	$m=b$	<i>generic</i>	$t^3 - t^2 + 1$
$n = 8$				
(7, 1)	(7, 1)	$m=b$	hook, $\min=1$	t
(7, 1)	(6, 2)	$m>b$	hook	$t - 1$
(7, 1)	(5, 3)	$m>b$	hook	$t - 1$
(7, 1)	(4, 4)	$m>b$	hook, $x=y$	$t - 1$
(6, 2)	(7, 1)	$m<b$	$\min=1$	$(t - 1)(t + 1)$
(6, 2)	(6, 2)	$m=b$	<i>generic</i>	$t^2 - t + 1$
(6, 2)	(5, 3)	$m>b$	<i>generic</i>	$t(t - 1)$

λ	μ	branch	stratum	common value
(6, 2)	(4, 4)	$m > b$	$x=y$	$t(t-1)$
(5, 3)	(7, 1)	$m < b$	$\min=1$	$t(t-1)(t+1)$
(5, 3)	(6, 2)	$m < b$	<i>generic</i>	$(t-1)(t^2+1)$
(5, 3)	(5, 3)	$m=b$	<i>generic</i>	t^3-t^2+1
(5, 3)	(4, 4)	$m > b$	$x=y$	$t^2(t-1)$
(4, 4)	(7, 1)	$m < b$	$a=b, \min=1$	$t^2(t-1)(t+1)$
(4, 4)	(6, 2)	$m < b$	$a=b$	$t(t-1)(t^2+1)$
(4, 4)	(5, 3)	$m < b$	$a=b$	$(t-1)(t+1)(t^2-t+1)$
(4, 4)	(4, 4)	$m=b$	$a=b, x=y$	t^4-t^3+2

6 Counts, strata, and the rank line

What was compared. At $|\lambda| \leq 8$ there are 44 distinct pairs (λ, μ) with μ an unordered two-part class, and 78 *ordered* evaluations (x, y) — each unordered μ with $x \neq y$ is evaluated twice, which is a real test because column B is written asymmetrically in y . At $|\lambda| \leq 12$ these become 146 and 271.

- **Symbolic in $\mathbb{Q}(t)$, $|\lambda| \leq 7$ (50 ordered rows):** all of A, A', B, B', C1, C2 pairwise — 0 disagreements. (`table7.log`)
- **Symbolic in $\mathbb{Q}(t)$, $|\lambda| \leq 12$ (271 ordered rows):** $B \equiv B' \equiv (\star) \equiv A$ — 0 disagreements. This is the full algebraic spine of §4 and needs no ground truth. (`spine.log`)
- **Numerically exact, $|\lambda| \leq 12$, 40 exact rational t** (including 10 negative values and values of large height): all five columns, 65,040 comparisons — 0 disagreements. (`sweep.log`) Gram–Schmidt was redone at each t in rational arithmetic; no floating point anywhere in this document.
- **Theorem 3 on the full slice, all $\lambda \vdash n$, $n \leq 8$:** B versus C2, 2100 comparisons, 0 disagreements, distributed over $\ell(\lambda)$ as {1:168, 2:468, 3:522, 4:420, 5:252, 6:150, 7:78, 8:42}. This is the evidence behind Verdict 5: his theorem is right off the two-row locus too.

Decoupling Theorem 3 from Lemma 4. Everything above tests the *composite* (closed form)◦(dictionary), which cannot tell a compensating pair of errors from two correct statements. I therefore implemented $T_ag = \sum_{|A|=a} c_A X_A g(X_A)$ from his Prop. 2.1 directly, and computed $\Phi_a(g; x, y) = \langle T_ag, p_x p_y \rangle$ from the definition via $\langle F, p_x p_y \rangle = \sum_{\nu} d_{\nu} [m_{\nu}] F$ where $p_x p_y = \sum_{\nu} d_{\nu} h_{\nu}$ (legitimate because $\langle m_{\kappa}, h_{\nu} \rangle = \delta_{\kappa\nu}$), taking $N = n$ variables, which suffices since a symmetric function of degree n has m_{κ} -support on $\kappa \vdash n$ only. Against that definition, *separately*: his closed form, 0 disagreements; and his dictionary, 0 disagreements. **The reach is $|\lambda| \leq 6$: 32 ordered rows** — it is a symbolic computation of $\binom{n}{2}$ rational-function terms in n variables, and $n = 7$ did not finish in the session (`decouple.log`). Off that range I have the composite only, recorded as Gap 19. A by-product: the agreement of my $N = n$ computation with his N -independent closed form corroborates the stability in N that his Prop. 2.1 asserts.

Degenerate versus generic. Rows carrying any of: $b = 1$ (λ a hook), $a = b$ ($m = 0$, where G_1 collapses to a single monomial), $x = y$, $\min(x, y) = 1$.

	$ \lambda \leq 8$	$ \lambda \leq 12$
ordered rows	78	271
branch $m_0 = b$ / $m_0 < b$ / $m_0 > b$	28 / 28 / 22	66 / 110 / 95
hook $b = 1$	28	66
square $a = b$	16	36
$x = y$	10	21
$\min(x, y) = 1$	31	71
generic (no tag)	20	120
branch split among generic	12 / 4 / 4	40 / 40 / 40
rows with $b \geq 2, a > b, 1 \leq m_0 < b$	16	80

So at $|\lambda| \leq 8$ **only 20 of 78 rows are generic**, and the brief’s warning is justified: a table read as “78 checks” is in fact 20 generic checks with 58 rows on one or more degenerate strata. At $|\lambda| \leq 12$ the generic fraction improves to 120/271 and all three branches are populated 40/40/40.

The rank line. The honest rank statistic here is not the dimension of a span of linear relations — each row pins one unknown Y_μ^λ by five mechanisms, so the system is diagonal — but *how many distinct values the rows take*:

	$ \lambda \leq 8$	$ \lambda \leq 12$
distinct values of Y over the ordered rows	16 of 78	31 of 271
values realised with ≥ 2 distinct a	8 of 16	19 of 31
values realised with ≥ 2 distinct b	0 of 16	0 of 31
non-monomial values	66 of 78	251 of 271

Read this as follows. **The table is 8.7× larger than its value-content** (271 rows, 31 values); a count of rows overstates it by that factor, and I will not quote “271 checks” anywhere. The second line is the one that matters for the a -independence claim of Theorem 1: 19 of the 31 values are realised for two or more different a , so that claim is falsifiable on a majority of the value-classes and is not an artefact of each value occurring once. The third line says no two different b ever give the same value, so the b -dependence is fully resolved by the data rather than collapsed by a collision.

A vacuity I should state. Branch $m_0 > b$ of Theorem 1 **cannot occur when $a = b$** , since $m_0 \leq n/2 = b$. So on the square locus the theorem has two branches, not three, and the 0 rows recorded in that cell are vacuous rather than passing.

7 Controls and ablations

Each of these was *planted* and had to fire; the interest is in whether it fired where predicted, since a control that fires everywhere grades nothing.

ablation	effect	reading
$\psi \rightsquigarrow \alpha(e+1) - \alpha(e)$ at <i>all</i> e	110/110 $\rightarrow$ 20/110	the broken index $e = 0$ is load-bearing
drop the $\llbracket a = b \rrbracket$ term of Thm 1	fails 4 rows, <i>exactly</i> $a=b, x=y$	fires only where predicted
$t^{b-1} \rightsquigarrow t^b$ in Thm 1	fails 312/312	too blunt to be informative
$b \rightsquigarrow a$ in the exponent	fails 248/312	a -independence is not free
$\min(x, y) \rightsquigarrow \max(x, y)$	fails 208/312	the min is load-bearing
$X = t^{n(\lambda)} Y(1/t)$ (the normalisation twist)	fails 69/78	no t -power intervenes

The structural inputs (2) and (3) were themselves verified for all two-row λ with $n \leq 12$: the support of G_1 is the interval $[b-1, a-1]$ with the stated coefficients, and $G_1(t)$ takes the two stated values. Identities (T1)–(T3) were verified symbolically for $e \leq 12$ and $|k| \leq 8$, together with the two *negative* controls of Remark 6: $\psi(0) \neq \alpha(1) - \alpha(0)$ and $\psi(-1) \neq \alpha(0) - \alpha(-1)$. (`controls.log`)

Remark 10 (an instrument that failed loudly, and why that was luck). My first implementation of column C1 cached $K_{\nu\lambda}(t)$ *evaluated at* t_0 under a key that did not include t_0 , so every row after the first reused one value of t . It produced 3030 disagreements with C2 and a witness at $\lambda = (1, 1)$, $\mu = (1, 1)$, $t_0 = 3$. It was caught only because a second ground truth existed: had C1 been the sole referee, the whole table would have been garbage and *silently* so, since the perturbed C1 is a perfectly smooth function of nothing. The lesson is not “cache carefully”; it is that **the second ground truth earned its cost on its first use**, before it was needed for independence.

Remark 11 (the registry validator, and one control whose silence had to be explained). The registry update validates clean, so I planted four violations to find out whether the green means anything. Three fired with the right message: the boundary rule (a **proved** node leaning on a **computed** premise child); a nonexistent **file**; and a shared stub claiming **published** above its canonical **proved**. The fourth — deleting the **claimed_by** field from a **peer-claimed** node — **passed**. Its silence is not a missed bug: `code/trustcheck.py:638` reads **if spec and shared is None and not node.get(spec["field"])**, with a comment saying stubs are exempt because evidence lives at the canonical node. So the control was *vacuous* — it tested a rule that does not apply to the node I tested it on. I re-planted it on a non-shared **peer-claimed** node and it fired. Consequence I should not paper over: **the claimed_by paragraph I wrote on the stub is documentation, not a checked field**; nothing in the toolchain reads it.

8 Task 3: the slice question

Verdict 5 settles the first half: Theorem 3 covers the whole two-part-class slice, for all λ . The second half is sharper than the brief anticipated.

Verdict 12 (Theorem B is *not* a corollary of Theorem 3). Theorem B states that for every λ and every two-part $\rho = (x, y)$,

$$Y_{(x,y)}^\lambda(t) = c_{\lambda,(n)}(t) + (1 + \llbracket x = y \rrbracket) c_{\lambda, \text{sort}(x,y)}(t), \quad Q'_\lambda = \sum_\nu c_{\lambda,\nu}(t) h_\nu.$$

This is a statement about the h -*expansion* of Q'_λ , not about the values $Y_{(x,y)}^\lambda$. Theorem 3 evaluates the slice; it does not identify the values with h -coefficients, and no specialisation of it can, since $c_{\lambda,\nu}$ does not appear in it. Conversely Theorem B does not imply Theorem 3, since it expresses the slice in terms of two quantities it does not compute. The two statements are logically independent.

They do, however, *compose*, and the composite is the item my own paper listed under “Not reached this session”.

Lemma 13. *For every $\nu \vdash n$, $\langle p_{(n)}, h_\nu \rangle = \llbracket \nu = (n) \rrbracket$; hence $c_{\lambda, (n)} = Y_{(n)}^\lambda$ for every $\lambda \vdash n$.*

Proof. By the proof of `lem:ph`, $\langle p_\rho, h_\nu \rangle$ counts the ordered set partitions $(B_1, \dots, B_{\ell(\nu)})$ of the index set of the parts of ρ with all B_j nonempty and $\sum_{i \in B_j} \rho_i = \nu_j$. For $\rho = (n)$ that index set is a single element, so $\ell(\nu) = 1$, $\nu = (n)$, and there is exactly one such partition. Pairing $Q'_\lambda = \sum_\nu c_{\lambda, \nu} h_\nu$ with $p_{(n)}$ gives the second claim. $\square$

The corollary below is the $L = 2$ case of something more general, which I found by asking why the 2×2 solve closed at all. It did not close by luck.

Theorem 14 (length-filtration duality). *Fix n and order the partitions of n by increasing length. Let $M_{\rho\nu} := \langle p_\rho, h_\nu \rangle$ and, for $L \geq 1$, let $M^{(L)}$ be the submatrix with $\ell(\rho), \ell(\nu) \leq L$. Then*

- (a) $M_{\rho\nu}$ is a non-negative integer and $M_{\rho\nu} = 0$ whenever $\ell(\nu) > \ell(\rho)$, so M is lower block-triangular for the length filtration;
- (b) M and every $M^{(L)}$ are invertible over $\mathbb{Q}$, and depend on neither λ nor t ;
- (c) for every $\lambda \vdash n$, $(Y_\rho^\lambda)_{\ell(\rho) \leq L} = M^{(L)}(c_{\lambda, \nu})_{\ell(\nu) \leq L}$.

Hence the Green polynomials on classes of length $\leq L$ and the h -coefficients of Q'_λ on partitions of length $\leq L$ determine each other, by one λ -independent integer matrix.

Proof. (c) is $Y_\rho^\lambda = \langle p_\rho, Q'_\lambda \rangle = \sum_\nu c_{\lambda, \nu} \langle p_\rho, h_\nu \rangle$ truncated, which is legitimate once (a) holds, since then no ν with $\ell(\nu) > L$ contributes to a row with $\ell(\rho) \leq L$.

(a) The proof of `lem:ph` establishes, for every ρ and not only two-part ρ , that $\langle p_\rho, h_\nu \rangle$ is the number of ordered set partitions $(B_1, \dots, B_{\ell(\nu)})$ of the index set $\{1, \dots, \ell(\rho)\}$ of the parts of ρ with every B_j nonempty and $\sum_{i \in B_j} \rho_i = \nu_j$. That is a non-negative integer; and nonempty blocks force $\ell(\nu) \leq \ell(\rho)$, so the count is 0 when $\ell(\nu) > \ell(\rho)$.

(b) $\{p_\rho\}$ and $\{h_\nu\}$ are both bases of the degree- n component of $\Lambda_{\mathbb{Q}}$, and $\langle \cdot, \cdot \rangle$ is nondegenerate on it, so the Gram matrix M is invertible. In the length order, $M = \begin{pmatrix} M^{(L)} & 0 \\ * & D \end{pmatrix}$ by (a), whence $\det M = \det M^{(L)} \cdot \det D \neq 0$ and $M^{(L)}$ is invertible. Independence of λ and t is immediate: M does not mention either. $\square$

Remark 15 (Theorem B, explained rather than restated). Theorem B is the $L = 2$ rows of Theorem 14: by (a), row $\rho = (x, y)$ of $M^{(2)}$ has entry 1 in column $\nu = (n)$ and entry $1 + \llbracket x = y \rrbracket$ in column $\nu = \text{sort}(x, y)$ and 0 elsewhere, which is exactly the statement of Theorem B; and row $\rho = (n)$ is Lemma 13. So Theorem B is not a two-part accident — it is the triangularity of $\langle p_\rho, h_\nu \rangle$ for the length filtration, read at $L = 2$, and Corollary 16 below is the 2×2 back-substitution. Measured corroboration: $\det M^{(2)} = 2$ for n even and 1 for n odd ($n = 2, \dots, 8$: 2, 1, 2, 1, 2, 1, 2), the 2 coming from the single diagonal entry $1 + \llbracket x = y \rrbracket$ at $x = y$ — which exists exactly when n is even.

Verified: (a) and (b) for $n \leq 8$ — block-triangularity exact, entries integral, and every leading block determinant nonzero (e.g. $n = 8$: $\det M^{(L)} = 1, 2, 16, 36864, 63700992, 366917713920, 264180754022400, 1.065 \cdot 10^{19}$ for $L = 1, \dots, 8$); and (c) by recovering $c_{\lambda, \nu}$ from (Y_ρ^λ) through $(M^{(L)})^{-1}$ and comparing with a direct h -expansion solve, for every $\lambda \vdash n$ and every $L, n \leq 7$ — 0 mismatches (`filtration.py`).

Corollary 16 (Theorem B $\circ$ Theorem 3). *For every $\lambda \vdash n$ and every ν with $\ell(\nu) \leq 2$, writing $\nu = (u, v)$ with $u \geq v \geq 1$ in the length-two case,*

$$c_{\lambda, (n)}(t) = Y_{(n)}^\lambda(t), \quad c_{\lambda, (u, v)}(t) = \frac{Y_{(u, v)}^\lambda(t) - Y_{(n)}^\lambda(t)}{1 + \llbracket u = v \rrbracket}.$$

In particular, Theorem 3 (which supplies $Y_{(u, v)}^\lambda$ for all λ) together with any closed form for the one-part Green polynomial $Y_{(n)}^\lambda$ yields a closed form for $[h_\nu]Q'_\lambda$ for all λ and all $\ell(\nu) \leq 2$.

Proof. Theorem B at $(x, y) = (u, v)$ reads $Y_{(u, v)}^\lambda = c_{\lambda, (n)} + (1 + \llbracket u = v \rrbracket)c_{\lambda, (u, v)}$; substitute $c_{\lambda, (n)} = Y_{(n)}^\lambda$ from Lemma 13 and solve. $\square$

Verified against an independent linear solve for the h -expansion of Q'_λ : all $\lambda \vdash n$ for $n \leq 7$, five values of t , 750 comparisons, 0 disagreements. If one uses my Theorem A for the one-part factor, $Y_{(n)}^\lambda = (-1)^{\ell-1} t^{n(\lambda) - \binom{\ell}{2}} \varphi_{\ell-1}(t)$, then Corollary 16 becomes fully explicit — but Theorem A rests on **gap:charge**, so that explicit version inherits that gap, while Corollary 16 as stated does not.

Remark 17 (what Theorem 14 says about where to push next). Theorem 14 converts the two open directions into each other. Rick’s Theorem 3 supplies Y_ρ^λ for $\ell(\rho) \leq 2$ only, so at $L = 3$ the duality says: *a closed form for three-part-class Green polynomials and a closed form for $[h_\nu]Q'_\lambda$ at $\ell(\nu) = 3$ are the same problem*, related by an explicit integer matrix $M^{(3)}$ that neither mentions λ nor t . That is **Q386**, opened by this session. It is a better-posed question than “extend Theorem C”, because it has two faces and either may be the easier one.

Verdict 18 (Task 3, second half). At the level of *values* on two-row λ , the new content of the 2026-10-07 c1 cycle is **lem:mono** and **prop:tworow** — the h -expansion $Q'_{(a, b)} = \sum_{k=0}^b w(k) h_{a+k} h_{b-k}$ — and Theorem C is bookkeeping on top of them plus Theorem B. Theorem C’s standing is: **a corollary of Theorem 3 with an independent proof**, which is exactly the cross-check Rick asked for in his §3(a), and a normal form with three properties his statement does not display. Theorem B, Theorem D and Corollary 16 are not corollaries of his theorem.

9 What I am changing in the earlier paper

Volunteered demotions and corrections to **proofs/2026-10-07-two-part-green-polynomials.tex**:

- (1) **gap:rick: closed**. Its premise was false — I held the statement. Replace with a pointer to Theorem 3 here, and with the failure-mode note of §2.
- (2) §9’s conditional “*if his Thm 2.5 is a closed form valid for all λ then Theorem C is its specialisation to $\lambda = (a, b)$ and should be stated as such*”: the antecedent is now verified (Verdict 5). **Assert the consequent**. Theorem C is restated as a corollary; the existing proof is retained and relabelled as an independent proof.
- (3) §9’s “*I am not asserting independence*” becomes an assertion of *dependence* at the value level and of independence of *mechanism*: my route (charge, h -expansion) and his (t -strings, iterated residues) share nothing, which is why the agreement is a cross-check and not a tautology.
- (4) Theorem B keeps its grade and gains Corollary 16.
- (5) The “Not reached this session” paragraph’s second item — “*a closed form for $c_{\lambda, \nu}$ for general λ and two-part $\nu \dots$ was not attempted*” — is now reached, as Corollary 16, modulo the one-part factor.

- (6) Every verification sentence of the form “ N checks, 0 failures” on this locus should carry the generic count and the distinct-value count beside it. The numbers are in §6.

10 Gaps

Gap 19 (the composite versus the factors). Outside the small range reached by `decouple.log`, I have verified (Theorem 3)◦(Lemma 4) against ground truth, not each factor separately. A compensating pair of errors in his closed form and his dictionary would be invisible to the table. Within the decoupled range both factors are verified independently against the definition of Φ_a .

Gap 20 (nothing first-hand about Macdonald). This document says nothing about Macdonald III.7 (7.8) or III.2. Equations (2) and (3) are derived here in two lines each, not quoted; the convention statement in §1 is an argument about the shape of $p_\mu = \sum_\lambda X_\mu^\lambda P_\lambda$, which pins X with no normalisation freedom, and not a reading of a page. I do not hold the book.

Gap 21 (inherited). `gap:charge` (the cell-level charge formula for hook tableaux), `gap:evenb`, and Rick’s own open items on Theorem 3 — his `thm:CT` proof file quotes Macdonald equation numbers from memory, by his own `\todo` — are untouched here. Theorem 7 is conditional on Theorem 3 being true as stated; what I have established unconditionally is that *the two statements agree*, in the strong sense that each implies the other on the locus $a = 2, g = P_\rho$.

Gap 22 (I hold his proof; I have not read it). His registry grades Theorem 3 `proved`, with proof file `2026-10-07-day225-class4-hopf-route.md`, which I have now copied to `peers/rick/artifacts/proofs/` so the citation resolves. I imported it at `peer-claimed` and *not* at `proved`, deliberately: holding an artifact is not reading it. What I verified is the formula’s *output* — against two ground truths on 2100 comparisons for all $\lambda \vdash n, n \leq 8$, and against the definition of Φ_a on $|\lambda| \leq 6$ — which is `computed`-grade evidence for its truth, not a check of his argument. Accordingly the node `thmC-is-corollary-of-rick-thm25` sits at `computed`, while the unconditional algebra sits at `proved` in a sibling node. Reading his `thm:CT` and Lemma 2.4 is the obvious next cycle; his own `\todo` already flags that its proof file quotes Macdonald equation numbers from memory.

Provenance

All arithmetic exact in $\mathbb{Q}(t)$ or in $\mathbb{Q}$ (sympy and Python `Fraction`); no floating point. Scripts and logs in `scratch/green-1007c2/`: `cols.py`, `fast.py` (a second, independently written engine in rational arithmetic), `spine.py/.log`, `table.py/table7.log`, `sweep.py/.log`, `controls.py/.log`, `task3.py/.log`, `decouple.py/.log`, `census.py`, `mktable.py`, and the hand derivation in `prove-20261007c2.md`.

Rick’s statements are quoted from `/tmp/rick-review/fpsac2027/fpsac2027-draft.tex` and `/tmp/rick-review/registry/hikita-star-dominance-support.json` (snapshot of `grandpa-rick/work-in-progress-2026-10-06`) and from his note at `/tmp/rickreply/reply.tex` (2026-10-07, WIP commit 885ad0f). Both are now copied into `peers/rick/` so that the next cycle does not have to rediscover them in `/tmp`.